

LRI Emergency Department

Acute aortic syndrome

Use in adults in whom AAS is the main concern or who have high-risk pain features or exam findings or risk factors for AAS listed in box 3 if presenting with:

- Chest, abdo or back pain
- Neurological deficit
- Mesenteric ischaemia
- Limb ischaemia
- Syncope
- BP differential >20
- Systolic blood pressure >180

Use individualized approach if pregnant, trauma patient or recent cocaine use

Disclaimer: This is a clinical template; clinicians should always use judgment when managing individual patients

Re-approved by EDGC on 25Feb26

Review due Feb31 - Trust Ref: C54/2021

Patient details

Full name

DoB

Unit number

(use sticker if available)

① Direct ECHO signs of AAS?

NB: Omit if timely ECHO expertise unavailable

Yes, at least one of the below

Intimal flap

Intramural aortic haematoma

Penetrating aortic ulcer

NO, none of the above

② Indirect ECHO signs of AAS?

NB: Omit if timely ECHO expertise unavailable

Yes, at least one of the below

Thoracic aorta dilatation (diameter ≥ 4 cm at any level)

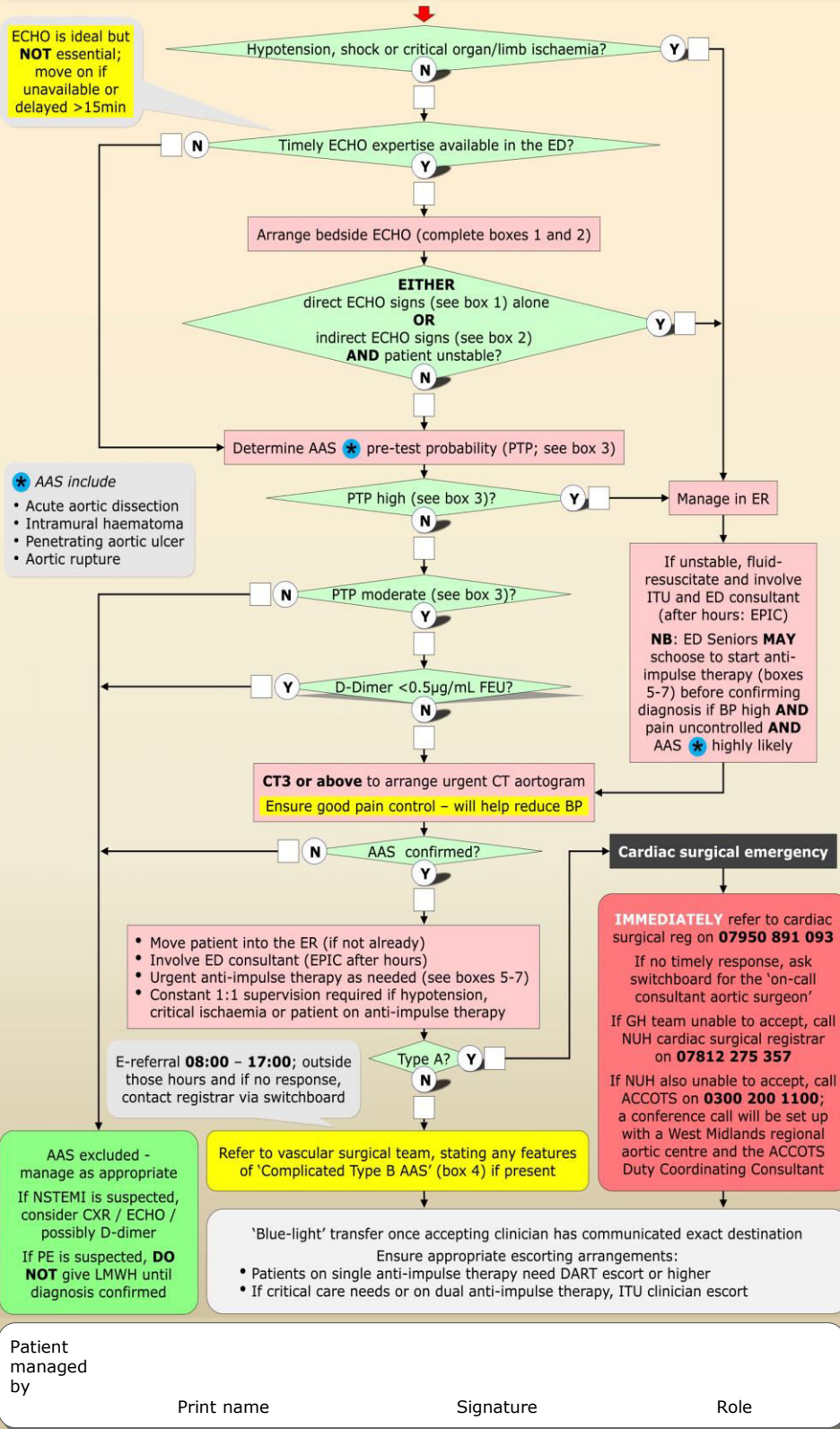
Pericardial effusion/tamponade

Aortic valve regurgitation at least

NO, none of the above

ECHO is ideal but NOT essential; move on if unavailable or delayed >15min

- AAS include
- Acute aortic dissection
 - Intramural haematoma
 - Penetrating aortic ulcer
 - Aortic rupture



③ AAS pre-test probability (PTP)

Tick applicable criteria & record final score

Known aortic aneurysm

2

Other risk factors

Marfan syndrome

Loeys-Dietz syndrome

Turner syndrome

Bicuspid aortic valve

Ehlers-Danlos Syndromes

Aortic valve disease

Aortic procedure within last 6 weeks (e.g. cardiac surgery, PCI, diagnostic coronary angiography, TAVI or TEVAR)

Family history of aortic dissection

Family history of intramural haematoma

Family history of ulcer with leak

Score if any of the above

1

High-risk chest/abdo/back pain qualities

Severe or worst ever

Thunderclap or abrupt

Tearing or ripping

Migrating or radiating

EITHER score if 1 or 2 of the above

OR score if 3 or 4 of the above

1

2

High-risk findings on assessment

Pulse deficit including limb ischaemia (check for decrease or absence of radial, carotid, subclavian and femoral pulses)

New neurological deficit:

- Stroke or TIA (most commonly RIGHT)
- Paraplegia
- Limb pain/paraesthesia/motor deficit
- Hypoxic encephalopathy (confusion)
- Horner syndrome

Aortic insufficiency / regurgitation

Hypotension

Pericardial effusion (see box 2) (omit if no ECHO expertise available)

Score if any of the above

2

Senior clinical judgement (ST3 or above)

Alternative diagnosis more likely

Unsure

AAS is most likely diagnosis

-1

0

1

2+ High

PTP is >5%

1 Moderate

PTP is 0.5-5%

0 Low

PTP is <0.5%

④ Complicated Type B AAS?

Yes, as at least one of the below

CT findings

Severe aortic dilatation

Aortic rupture signs

Haemothorax

Periaortic/mediastinal haematoma

Organ malperfusion features

AKI

Metabolic acidosis

Rising lactate

Lower limb ischaemia

Paraplegia

Leg pain/paraesthesia/motor deficit

NO, as none of the above

⑤ Esmolol infusion

• Our pre-mixed esmolol infusion bags contain 2.5G in 250mL of 0.9% NaCl (10mg/mL)

• Tick nearest weight in table below; the corresponding column shows the infusion rates needed

• In NC Meds, go to Emergency Medicine (ED) > Vasoactive meds (ED) > Esmolol. Select nearest wt and prescribe '60-sec bolus & maintenance infusion' + 'extra 60-sec boluses' (both are preselected).

• If NC is offline, prescribe infusion on a paper drug chart as per box 6 below

• **NB:** Infusion start and rate changes are best captured initially as verbal orders on the 'ER critical interventions log' rather than trying to record them in NC Meds in real time

				Patient weight (in kg ; select nearest value)												
				50	55	60	65	70	75	80	85	90	95	100	105	>109
Dose in microgram/kg/min	500	...delivered by setting infusion rate (in mL/h) to	60-sec bolus	150	165	180	195	210	225	240	255	270	285	300	315	330
	50		Maintenance	15	16.5	18	19.5	21	22.5	24	25.5	27	28.5	30	31.5	33
	100			30	33.0	36	39.0	42	45.0	48	51.0	54	57.0	60	63.0	66
	150			45	49.5	54	58.5	63	67.5	72	76.5	81	85.5	90	94.5	99
	200			60	66.0	72	78.0	84	90.0	96	102.0	108	114.0	120	126.0	132
	250			75	82.5	90	97.5	105	112.5	120	127.5	135	142.5	150	157.5	165
	300			90	99.0	108	117.0	126	135.0	144	153.0	162	171.0	180	189.0	198

⑥ Esmolol example prescription during Nervecentre downtime

PARENTERAL INFUSIONS

Infusion Fluid		Additions to Infusion					
Date	Type/Strength	Vol.	Medicine	Dose	Route	Time to run or ml/hr	Prescriber
DD/MM/YY	10mg/mL	250mL	Esmolol (premixed)	2.5g	IV	255 mL/h 60-second bolus only	YOUR NAME & signature
DD/MM/YY	10mg/mL	250mL	Esmolol (premixed)	2.5g	IV	25.5 - 153mL/h	NAME & sig

⑦ Anti-impulse therapy

• IV-trained nurse competent to manage fluid infusion pump required

• Initial titration phase requires concentration – clinician & nurse must stay with patient throughout

• Take this proforma to the bedside to guide you through the infusion rate changes & other actions

• **NB:** Sudden or accidental infusion stops cause hazardous BP spikes – **protect IV access**

• Consider inserting arterial line to allow accurate, real-time BP monitoring

• **NB:** DO NOT insert arterial line into a limb with ischemia / pulse deficit

SBP > 120?

Y

N

HR > 60?

Y

N

• Prescribe esmolol infusion on NC Meds as per box 5

• Begin with **500** microgram/kg/min for **60sec only** then reduce to **50mcg/kg/min**

• Check HR and BP @ **4min**

if HR > 60 AND SBP > 120

• Increase to **500** microgram/kg/min for **60sec only** then reduce to **100mcg/kg/min**

• Check HR and BP @ **4min**

if HR > 60 AND SBP > 120

• Increase to **500** microgram/kg/min for **60sec only** then reduce to **150mcg/kg/min**

• Check HR and BP @ **4min**

if HR > 60 AND SBP > 120

• Increase to **500** microgram/kg/min for **60sec only** then reduce to **200mcg/kg/min**

• Check HR and BP @ **4min**

if HR > 60 AND SBP > 120

• Increase to **500** microgram/kg/min for **60sec only** then reduce to **250mcg/kg/min**

• Check HR and BP @ **4min**

if HR > 60 AND SBP > 120

• Increase to **500** microgram/kg/min for **60sec only** then reduce to **300mcg/kg/min**

• Check HR and BP @ **4min**

if HR > 60 AND SBP > 120

Reduce rate by 50mcg/kg/min

SBP < 100?

Y

N

SBP < 120?

Y

N

Continue esmolol infusion at present rate

ED senior doctor or ICU clinician to administer only

• Continue esmolol infusion at present rate

• In NC Meds, go to 'Emergency Medicine (ED) > Prescribe & administer (ED) > GTN bolus injection'; each dose to be prescribed separately

• Draw up 10mL of neat GTN (50mg in 50mL vial) in a 10mL syringe

• Administer 0.5-2mg bolus IV (depending on how high SBP still is)

• Record BP **every 5min** and give a further dose whenever SBP > 120

HR

SBP

HR

SBP

HR

SBP

HR

SBP

HR

SBP

HR

SBP

HR

SBP

(example for 85kg pt as per table in Box 5):

Start infusion at 255mL/h; after 60sec reduce to 25.5mL/h)

(example for 85kg pt as per table in Box 5):

Increase again to 255mL/h; after 60sec reduce to 51.0mL/h)